OAVS TGT COMPUTER SCIENCE TEACHER EXAMINATION

MCQ QUESTION BANK

Chapter 3 • Hierarchical Database

3.1 Introduction to Hierarchical Database

150 Unique, Exam-Focused Multiple Choice Questions with Answers & Revision-Oriented Explanations

Basic • Definition • Concept • Feature • Statement • Assertion-Reason • Match-the-Following • Comparison • Scenario • Application • Previous-Year Pattern • Difficult • Tricky

Table of Contents

150 Questions

OAVS TGT Computer Science — Hierarchical Database Syllabus

Chapter 3: Hierarchical Database 3

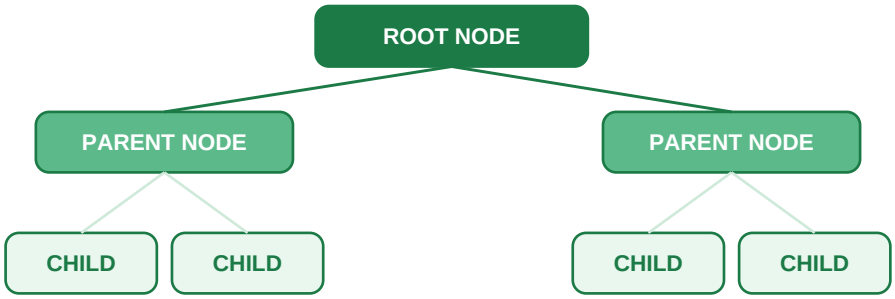
3.1.1 Basic Concept (9 MCQs) . . . 3
3.1.2 Definition (9 MCQs) . . . 6
3.1.3 History (8 MCQs) . . . 9
3.1.4 Characteristics (10 MCQs) . . . 12
3.1.5 Tree Structure (10 MCQs) . . . 15
3.1.6 Root Node (7 MCQs) . . . 18
3.1.7 Parent Node (7 MCQs) . . . 21
3.1.8 Child Node (7 MCQs) . . . 23
3.1.9 Levels (8 MCQs) . . . 25
3.1.10 Parent-Child Relationship (9 MCQs) . . . 28
3.1.11 One-to-Many Relationship (8 MCQs) . . . 31
3.1.12 Data Organization (9 MCQs) . . . 34
3.1.13 Hierarchical Database Model (7 MCQs) . . . 37
3.1.14 Navigation (7 MCQs) . . . 39
3.1.15 Advantages (9 MCQs) . . . 41
3.1.16 Limitations (9 MCQs) . . . 44
3.1.17 Applications (9 MCQs) . . . 47
3.1.18 Comparison with File Organization (8 MCQs) . . . 50

Chapter 3: Hierarchical Database

150 Questions

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A **Hierarchical Database** organizes data as a collection of records connected in a tree structure, where every child record is linked to exactly one parent record, forming a **one-to-many relationship**. The figure below illustrates the Root, Parent, and Child levels referenced throughout this question bank.



3.1.1 Basic Concept

9 Questions

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Q1 | Basic | Basic Concept

A Hierarchical Database organizes data primarily in the form of a:

(A) Flat table

(B) Network of nodes

(C) Random list

(D) Tree structure

Correct Answer: D

Explanation: Hierarchical databases arrange records in a tree-like structure with parent and child nodes.

Q2 | Definition-based | Basic Concept

Which of the following best defines a Hierarchical Database?

A. A database model that stores only unstructured text data

B. A database model that connects any two records directly

C. A database model that organizes data in a tree-like structure with parent-child relationships

D. A database model that stores data in unrelated rows and columns

Correct Answer: C

Explanation: A Hierarchical Database is defined by its tree-shaped organization where each record is linked to a single parent.

Q3 | Concept-based | Basic Concept

In a Hierarchical Database, data is logically arranged as a collection of:

A. Independent files with no links

B. Unordered document collections

C. Random key-value pairs

D. Records connected through parent-child links

Correct Answer: D

Explanation: The hierarchical model links records through parent-child associations, forming a structured tree.

Q4 | Basic | Basic Concept

The Hierarchical Database model is considered one of the:

A. Newest NoSQL database models

B. Cloud-native database models

C. Oldest database models used in DBMS

D. Graph-based database models

Correct Answer: C

Explanation: The hierarchical model was among the earliest database models developed before relational databases.

Q5 | Feature-based | Basic Concept

Which feature distinguishes a Hierarchical Database from a simple file system?

A. Storage in plain text files only
B. Absence of any relationship between records
C. Use of spreadsheets for storage
D. Defined parent-child relationships between records

Correct Answer: D

Explanation: Unlike plain file systems, hierarchical databases explicitly define parent-child relationships between records.

Q6 | Tricky | Basic Concept

A Hierarchical Database can best be visualized as resembling which real-world structure?

A. A social media friend network
B. A spreadsheet grid
C. A road map with multiple intersections
D. A family tree / organizational chart

Correct Answer: D

Explanation: A family tree or organizational chart mirrors the parent-child, one-to-many nature of hierarchical data.

Q7 | Statement-based | Basic Concept

Which statement about the Hierarchical Database is CORRECT?

A. Records exist without any hierarchy
B. Every child record can have multiple parent records
C. Every child record is linked to exactly one parent record
D. All records are stored at the same level

Correct Answer: C

Explanation: A core rule of the hierarchical model is that each child has only one parent, unlike network databases.

Q8 | Assertion-Reason | Basic Concept

Assertion (A): Hierarchical Databases are efficient for representing one-to-many relationships.
Reason (R): Data in a hierarchical database is structured as a tree with parent-child links.
Choose the correct option:

(A) Both A and R are true, but R does not explain A

(B) A is false, R is true

(C) Both A and R are true, and R correctly explains A

(D) A is true, R is false

Correct Answer: C

Explanation: The tree structure with parent-child links is exactly why hierarchical databases suit one-to-many data.

Q9 | Previous-Year Pattern | Basic Concept

In DBMS terminology, the Hierarchical Database model is classified as a:

(A) Record-based logical database model

(B) Physical storage model only

(C) Object-based database model

(D) Query language model

Correct Answer: A

Explanation: The hierarchical model is a record-based logical model describing how records relate, not a physical storage scheme.

3.1.2 Definition

9 Questions

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Q10 | Definition-based | Definition

A Hierarchical Database can be defined as a data model where each record has:

(A) No parent and no children

(B) Many parents but only one child

(C) One parent, but may have many children

(D) Many parents and many children

Correct Answer: C

Explanation: By definition, hierarchical records follow a one-parent, many-children structure.

Q11 | Definition-based | Definition

The term 'hierarchical' in Hierarchical Database refers to:

(A) Data arranged by file size

(B) Data arranged alphabetically

(C) Data arranged randomly across servers

(D) Data arranged in ranked levels from general to specific

Correct Answer: D

Explanation: 'Hierarchical' implies ranked levels, moving from broader (parent) to more specific (child) data.

Q12 | Concept-based | Definition

Which of the following is the most accurate technical definition of the hierarchical model?

(A) A model representing data as key-value pairs without structure

(B) A model representing data as graphs with many-to-many links

(C) A model representing data as independent tables joined by foreign keys

(D) A model representing data as a collection of records connected by one-to-many parent-child links, forming a tree

Correct Answer: D

Explanation: This definition captures the tree-based, one-to-many, parent-child nature unique to hierarchical databases.

Q13 | Basic | Definition

In simple terms, a Hierarchical Database stores data:

(A) In a two-dimensional grid only

(B) In multiple independent trees with no root

(C) In an inverted tree with a single root at the top

(D) In a circular chain of records

Correct Answer: C

Explanation: The hierarchical structure is often depicted as an inverted tree, starting from one root node.

Q14 | Statement-based | Definition

Which of the following statements correctly completes the definition: 'A Hierarchical Database model organizes records...'?

A. ...in graphs where nodes can have multiple parents

B. ...without any defined structure

C. ...in tables joined using primary and foreign keys

D. ...in a tree structure where each child has a single parent

Correct Answer: D

Explanation: This is the standard definition distinguishing the hierarchical model from relational and network models.

Q15 | Tricky | Definition

Which term is NOT part of the standard definition of a Hierarchical Database?

A. Foreign key constraint

B. Parent-child relationship

C. Root node

D. Tree structure

Correct Answer: A

Explanation: Foreign keys are associated with the relational model, not the hierarchical model's definition.

Q16 | Feature-based | Definition

The defining relationship type in a Hierarchical Database is:

A. Zero-to-zero

B. One-to-many

C. One-to-one only

D. Many-to-many

Correct Answer: B

Explanation: Hierarchical databases are fundamentally defined by their one-to-many parent-child relationships.

Q17 | Comparison-based | Definition

How does the definition of a Hierarchical Database differ from that of a Network Database?

(A) A hierarchical record can have multiple parents, while a network record has only one

(B) Both restrict records to zero parents

(C) Both allow unlimited parents for each record

(D) A hierarchical record has only one parent, while a network record can have multiple parents

Correct Answer: D

Explanation: The key definitional difference is that hierarchical models restrict each child to a single parent.

Q18 | Application-based | Definition

If a company wants to define departments and employees such that each employee reports to exactly one department, which model's definition fits best?

(A) Network Database model

(B) Flat file with no relationships

(C) Hierarchical Database model

(D) Distributed file model

Correct Answer: C

Explanation: The one-parent-per-child restriction matches the defining property of the hierarchical model.

3.1.3 History

8 Questions

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Q19 | Basic | History

The Hierarchical Database model became widely used starting in which decade?

(A) 2000s

(B) 1990s

(C) 1980s

(D) 1960s

Correct Answer: D

Explanation: Hierarchical databases gained prominence in the 1960s, notably through IBM's IMS system.

Q20 | Previous-Year Pattern | History

Which early IBM product is historically associated with the Hierarchical Database model?

(A) IMS (Information Management System)

(B) DB2

(C) Oracle RDBMS

(D) MySQL

Correct Answer: A

Explanation: IBM's IMS, developed in the 1960s, is the classic example of a hierarchical database system.

Q21 | Concept-based | History

The Hierarchical Database model was developed primarily to manage data for which type of early project?

(A) Large-scale industrial and government record-keeping projects

(B) Web-based social networks

(C) Cloud gaming platforms

(D) Personal mobile applications

Correct Answer: A

Explanation: Hierarchical databases originated to handle large structured records in industrial and government systems.

Q22 | Statement-based | History

Which statement about the origin of hierarchical databases is TRUE?

(A) The hierarchical model predates the relational database model

(B) The hierarchical model was created for graph analytics

(C) The hierarchical model originated from NoSQL databases

(D) The hierarchical model was developed after the relational model

Correct Answer: A

Explanation: Hierarchical databases existed before Codd introduced the relational model in 1970.

Q23 | Tricky | History

Which of these database models chronologically came FIRST?

A. Relational Database model
B. NoSQL Document model
C. Object-Oriented Database model
D. Hierarchical Database model

Correct Answer: D

Explanation: The hierarchical model is one of the oldest, predating relational, object-oriented, and NoSQL models.

Q24 | Basic | History

Why did early hierarchical databases suit mainframe computers of the 1960s-70s?

A. They automatically handled unstructured web data
B. They were designed for distributed cloud servers
C. They allowed fast, structured access to large volumes of predictable data
D. They required no storage hardware

Correct Answer: C

Explanation: Mainframes needed efficient structured access, which the tree-based hierarchical model provided.

Q25 | Comparison-based | History

Compared to later models, the historical hierarchical model was primarily limited by:

A. Its reliance on internet connectivity
B. Its rigid, single-parent tree structure
C. Its excessive flexibility in relationships
D. Its use of unstructured data only

Correct Answer: B

Explanation: Its rigid tree structure, while efficient, later proved limiting compared to relational flexibility.

Q26 | Application-based | History

A historian studying early banking record systems finds tree-based record files from the 1960s. These most likely used:

(A) A cloud-based NoSQL model

(B) An object-relational model

(C) A Hierarchical Database model

(D) A distributed blockchain model

Correct Answer: C

Explanation: Tree-based record files from that era are characteristic of the hierarchical database model.

3.1.4 Characteristics

10 Questions

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Q27 | Feature-based | Characteristics

Which of the following is a key characteristic of the Hierarchical Database model?

(A) All nodes are at the same level

(B) Nodes exist without any connection

(C) Each child node can have multiple parent nodes

(D) Each child node has exactly one parent node

Correct Answer: D

Explanation: A defining characteristic is the strict one-parent rule for every child node.

Q28 | Feature-based | Characteristics

Which characteristic makes navigation predictable in a Hierarchical Database?

(A) Random connections between unrelated nodes

(B) Multiple roots without hierarchy

(C) A fixed, well-defined tree path from root to any node

(D) Absence of any starting point

Correct Answer: C

Explanation: Because paths follow a fixed tree structure from the root, navigation is predictable.

Q29 | Statement-based | Characteristics

Which of the following is NOT a characteristic of a Hierarchical Database?

(A) Data follows a top-down structure

(B) Records are organized as a tree

(C) Records can have multiple parent records

(D) Each record has one parent

Correct Answer: C

Explanation: Allowing multiple parents per record is a network database trait, not a hierarchical one.

Q30 | Concept-based | Characteristics

The 'top-down' characteristic of a Hierarchical Database means data flows from:

(A) General/root-level information to specific/detailed child information

(B) Specific child data to general root data

(C) Random nodes to other random nodes

(D) External files into unrelated tables

Correct Answer: A

Explanation: Hierarchical data is organized top-down, starting broad at the root and narrowing at child levels.

Q31 | Assertion-Reason | Characteristics

Assertion (A): Hierarchical Databases enforce strict structural rules.
Reason (R): Every child record must be linked to only one parent record.
Choose the correct option:

(A) A is false, but R is true

(B) Both A and R are true, and R correctly explains A

(C) Both A and R are false

(D) A is true, but R is false

Correct Answer: B

Explanation: The single-parent rule is precisely what enforces the strict structural discipline in the model.

Q32 | Feature-based | Characteristics

Which characteristic helps ensure fast retrieval of related records in a Hierarchical Database?

A. Predefined parent-child access paths
B. Storing data without relationships
C. Absence of any access path
D. Unstructured scattering of data

Correct Answer: A

Explanation: Predefined access paths along the tree allow quick traversal to related records.

Q33 | Comparison-based | Characteristics

Compared to a flat file, a key characteristic advantage of hierarchical organization is:

A. Explicit relationships between records via parent-child links
B. Complete absence of any relationships
C. Storage of data in a single unstructured block
D. Random ordering of records

Correct Answer: A

Explanation: Flat files lack explicit relationships, while hierarchical databases define them clearly.

Q34 | Tricky | Characteristics

Which characteristic can become a drawback as data complexity increases?

A. Flexible many-to-many linking
B. Unlimited parent connections
C. Rigid one-parent restriction limiting relationship types
D. Automatic schema-less storage

Correct Answer: C

Explanation: The strict one-parent rule, while simple, restricts modeling of more complex relationships.

Q35 | Scenario-based | Characteristics

A school database stores 'Class' as parent and 'Student' as child, where each student belongs to only one class. This scenario reflects which characteristic?

(A) Many-to-many relationship

(B) No relationship between entities

(C) Strict one-to-many parent-child relationship

(D) Circular relationship

Correct Answer: C

Explanation: Each student linked to exactly one class is a textbook example of the one-to-many hierarchical trait.

Q36 | Basic | Characteristics

A common characteristic of hierarchical databases regarding data redundancy is that they may show increased redundancy when:

(A) Data is organized strictly under one parent

(B) No child records exist

(C) Only the root node stores data

(D) The same child data needs to relate to more than one parent

Correct Answer: D

Explanation: Since a child can have only one parent, representing shared data under multiple parents causes duplication.

3.1.5 Tree Structure — 10 Questions

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Q37 | Basic | Tree Structure

In the tree structure of a Hierarchical Database, the topmost node is called the:

(A) Leaf node

(B) Root node

(C) Child node

(D) Sibling node

Correct Answer: B

Explanation: The root node sits at the top of the tree and has no parent of its own.

Q38 | Concept-based | Tree Structure

In a hierarchical tree structure, nodes with no further children are called:

(A) Leaf nodes
(B) Branch nodes
(C) Root nodes
(D) Index nodes

Correct Answer: A

Explanation: Leaf nodes are the terminal nodes at the bottom of the tree with no children beneath them.

Q39 | Feature-based | Tree Structure

In the tree structure, connections between parent and child nodes are called:

(A) Keys
(B) Edges or links
(C) Indexes
(D) Attributes

Correct Answer: B

Explanation: The lines connecting parent and child nodes in a tree diagram are referred to as edges or links.

Q40 | Statement-based | Tree Structure

Which statement about the tree structure of a Hierarchical Database is TRUE?

(A) A tree structure allows cycles between nodes
(B) A tree structure can have multiple root nodes
(C) A tree structure has no leaf nodes
(D) A tree structure has exactly one root node

Correct Answer: D

Explanation: A valid hierarchical tree has a single root node from which all other nodes descend.

Q41 | Comparison-based | Tree Structure

How does a hierarchical tree structure differ from a general graph structure?

(A) Both structures are identical

(B) A graph structure has only one parent per node

(C) A tree structure has no cycles and each node has only one parent

(D) A tree structure allows multiple parents per node

Correct Answer: C

Explanation: Trees are acyclic with single parentage, while graphs can have cycles and multiple connections.

Q42 | Scenario-based | Tree Structure

Consider a tree: Company -> Department -> Employee. Which node is the leaf?

(A) Department

(B) Company

(C) Employee

(D) None of these

Correct Answer: C

Explanation: Employee is at the bottom of the hierarchy with no further children, making it the leaf node.

Q43 | Tricky | Tree Structure

In a hierarchical tree, a node that is both a child of one node and a parent of another is called:

(A) An orphan node

(B) A root node

(C) A leaf node

(D) An intermediate (or non-leaf, non-root) node

Correct Answer: D

Explanation: Such nodes exist between the root and leaves, acting as both child and parent simultaneously.

Q44 | Application-based | Tree Structure

A university database has Faculty -> Department -> Course. If 'Faculty' is the root, what is 'Department'?

(A) Unrelated to Faculty

(B) A child of Faculty and parent of Course

(C) A leaf node with no parent

(D) The root node

Correct Answer: B

Explanation: Department sits between Faculty (parent) and Course (child), making it an intermediate node.

Q45 | Match-the-Following | Tree Structure

Match the tree term with its description:
1. Root 2. Leaf 3. Edge 4. Node
A. Connection between records B. Topmost record C. Any record in the tree D. Record with no children

(A) 1-A, 2-B, 3-C, 4-D

(B) 1-B, 2-D, 3-A, 4-C

(C) 1-C, 2-A, 3-D, 4-B

(D) 1-D, 2-C, 3-B, 4-A

Correct Answer: B

Explanation: Root=topmost record, Leaf=record with no children, Edge=connection, Node=any record in the tree.

Q46 | Basic | Tree Structure

The overall shape formed by parent and child nodes together in a Hierarchical Database is known as a(n):

(A) Ring

(B) Tree (hierarchical structure)

(C) Star topology

(D) Matrix

Correct Answer: B

Explanation: The combined arrangement of root, intermediate, and leaf nodes forms the characteristic tree shape.

3.1.6 Root Node

7 Questions

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Q47 | Definition-based | Root Node

The root node in a Hierarchical Database is defined as:

(A) A node with no children

(B) Any node with multiple children

(C) The last node in the hierarchy

(D) The single node at the top of the hierarchy with no parent

Correct Answer: D

Explanation: The root is unique because it has no parent above it and starts the entire hierarchy.

Q48 | Feature-based | Root Node

How many root nodes can exist in a single valid hierarchical tree?

(A) Two

(B) As many as needed

(C) Exactly one

(D) Zero

Correct Answer: C

Explanation: A properly structured hierarchical tree must have exactly one root node.

Q49 | Concept-based | Root Node

All access to data in a Hierarchical Database typically begins at the:

(A) Sibling node

(B) Root node

(C) Leaf node

(D) Any random node

Correct Answer: B

Explanation: Traversal in hierarchical databases conventionally starts from the root and moves downward.

Q50 | Statement-based | Root Node

Which of these statements about the root node is FALSE?

A. The root node is the topmost node

B. The root node has no parent

C. The root node can have multiple children

D. The root node must have a parent node above it

Correct Answer: D

Explanation: By definition, the root has no parent; stating that it must have one is incorrect.

Q51 | Scenario-based | Root Node

In a hierarchy: Organization -> Branch -> Employee, which node is the root?

A. None of these

B. Branch

C. Employee

D. Organization

Correct Answer: D

Explanation: Organization sits at the top with no parent above it, making it the root node.

Q52 | Tricky | Root Node

If a hierarchical structure appears to have two nodes with no parent, what does this indicate?

A. It is a normal, valid hierarchical tree

B. It means all nodes are leaves

C. It means the database has no data

D. It represents two separate trees, not one valid single-root hierarchy

Correct Answer: D

Explanation: A true single hierarchical tree must have only one root; two parentless nodes suggest separate trees.

Q53 | Application-based | Root Node

In a hospital hierarchical database with Hospital -> Department -> Doctor -> Patient, the root node stores information about the:

(A) Hospital

(B) Patient

(C) Doctor

(D) Department

Correct Answer: A

Explanation: Hospital is the topmost entity in this hierarchy, making it the root node.

3.1.7 Parent Node

7 Questions

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Q54 | Definition-based | Parent Node

A parent node in a Hierarchical Database is defined as a node that:

(A) Has one or more child nodes linked beneath it

(B) Has multiple parent nodes above it

(C) Exists independently with no links

(D) Has no children at all

Correct Answer: A

Explanation: A parent node is characterized by having one or more child nodes connected below it.

Q55 | Feature-based | Parent Node

Can a single parent node have more than one child node in a Hierarchical Database?

(A) Only the root can have children

(B) Yes, a parent can have multiple children

(C) No, parents cannot have children

(D) No, a parent can have only one child

Correct Answer: B

Explanation: The one-to-many relationship allows any parent node to have multiple child nodes.

Q56 | Concept-based | Parent Node

Which of the following best describes the role of a parent node?

A. It represents a broader category that groups related child records
B. It only exists at the bottom of the tree
C. It stores unrelated random data
D. It always has zero children

Correct Answer: A

Explanation: Parent nodes represent broader categories under which more specific child data is grouped.

Q57 | Statement-based | Parent Node

Which statement about parent nodes is TRUE in a Hierarchical Database?

A. A parent node can also be a child of another node higher in the tree
B. A parent node can never be a child of another node
C. A parent node must always be the root
D. A parent node cannot have any children

Correct Answer: A

Explanation: Except for the root, every parent node in the tree can itself be a child of a higher-level node.

Q58 | Scenario-based | Parent Node

In hierarchy: Library -> Section -> Book, which node acts as a parent to 'Book'?

A. Library
B. Book itself
C. None of these
D. Section

Correct Answer: D

Explanation: Section is directly above Book in the hierarchy, making it the immediate parent of Book.

Q59 | Tricky | Parent Node

Can a parent node in a Hierarchical Database exist without any child nodes at a given time?

(A) No, parent nodes are never allowed to be empty

(B) Only leaf nodes can lack children

(C) No, a parent node must always have children

(D) Yes, it can exist without children until child records are added

Correct Answer: D

Explanation: A node is only classified as a parent when children exist; it can also temporarily hold zero children.

Q60 | Application-based | Parent Node

In an e-commerce hierarchical database with Category -> Sub-category -> Product, which node is the parent of 'Sub-category'?

(A) Customer

(B) Category

(C) Sub-category itself

(D) Product

Correct Answer: B

Explanation: Category directly precedes Sub-category in the hierarchy, making it its parent node.

3.1.8 Child Node

7 Questions

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Q61 | Definition-based | Child Node

A child node in a Hierarchical Database is defined as a node that:

(A) Is linked beneath multiple parent nodes

(B) Has no parent at all

(C) Always exists at the root level

(D) Is linked beneath exactly one parent node

Correct Answer: D

Explanation: A child node's defining trait is its single connection to one parent node above it.

Q62 | Feature-based | Child Node

Can a child node also serve as a parent node to other records?

A. Yes, but only if it has two parents
B. No, only the root can be a parent
C. Yes, unless it is a leaf node
D. No, a child node can never be a parent

Correct Answer: C

Explanation: Non-leaf child nodes can have their own children, making them parents at their level too.

Q63 | Statement-based | Child Node

Which of the following is TRUE regarding child nodes?

A. Each child node can be connected to multiple parent nodes
B. Child nodes always appear at the top of the tree
C. Each child node is connected to only one parent node
D. Child nodes cannot have their own children

Correct Answer: C

Explanation: The single-parent restriction for child nodes is a core rule of the hierarchical model.

Q64 | Scenario-based | Child Node

In hierarchy: School -> Class -> Student, identify the child node of 'Class'.

A. Student
B. School
C. Teacher
D. Class itself

Correct Answer: A

Explanation: Student is directly linked beneath Class, making it the child node of Class.

Q65 | Concept-based | Child Node

What happens to a child node if its parent node is deleted, in a strict hierarchical structure?

A. Nothing changes for the child node

B. The child node typically becomes inaccessible or is also removed

C. The child node gains multiple new parents

D. The child node automatically becomes the new root

Correct Answer: B

Explanation: Since access paths run through the parent, removing the parent affects the connected child node.

Q66 | Tricky | Child Node

If Node X has two parent nodes pointing to it, is X a valid child node in a Hierarchical Database?

A. Yes, but only in the root level

B. Yes, this is a normal hierarchical child node

C. No, this violates the single-parent rule of the hierarchical model

D. It depends on the number of grandchildren

Correct Answer: C

Explanation: A node with two parents fits a network database, not the strict hierarchical single-parent model.

Q67 | Application-based | Child Node

In a hierarchical database: Country -> State -> City, the child node of 'State' is:

A. Country

B. State itself

C. City

D. Continent

Correct Answer: C

Explanation: City is directly nested beneath State, making it the child node relative to State.

3.1.9 Levels

8 Questions

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Q68 | Definition-based | Levels

In a Hierarchical Database, a 'level' refers to:

(A) The color coding of nodes

(B) The total storage size of a node

(C) The number of columns in a table

(D) The depth or rank of a node measured from the root

Correct Answer: D

Explanation: Levels indicate how far a node is positioned from the root along the tree's depth.

Q69 | Basic | Levels

The root node is generally considered to be at which level?

(A) Level 0 (or Level 1, depending on convention)

(B) Undefined level

(C) The deepest level of the tree

(D) A negative level

Correct Answer: A

Explanation: By convention, the root node occupies the topmost level, often labeled Level 0 or Level 1.

Q70 | Concept-based | Levels

As you move down levels in a Hierarchical Database, the data generally becomes:

(A) More general and broad

(B) More specific and detailed

(C) Completely unrelated

(D) Random and unordered

Correct Answer: B

Explanation: Each descending level typically represents more specific data than the level above it.

Q71 | Scenario-based | Levels

In hierarchy: Institute (Level 1) -> Department (Level 2) -> Teacher (Level 3), which level does 'Teacher' belong to?

(A) Level 2

(B) Level 1

(C) Level 0

(D) Level 3

Correct Answer: D

Explanation: Teacher is the third tier down from the root Institute, placing it at Level 3.

Q72 | Statement-based | Levels

Which statement about levels in a Hierarchical Database is TRUE?

(A) All nodes at the same level must share one parent

(B) Only the root can occupy a level

(C) Nodes at the same level can have different parents

(D) Levels have no bearing on data organization

Correct Answer: C

Explanation: Sibling groups under different parents can still occupy the same overall level of the tree.

Q73 | Feature-based | Levels

What determines the number of levels in a given Hierarchical Database?

(A) The physical size of the storage disk

(B) The programming language used

(C) The depth of the data hierarchy being modeled

(D) The number of users accessing the database

Correct Answer: C

Explanation: The number of levels depends on how many layers of parent-child relationships the data requires.

Q74 | Tricky | Levels

Two nodes exist at the same level but under different parent nodes. Are they automatically related to each other?

(A) Yes, they automatically share the same parent

(B) No, being at the same level does not imply a direct relationship

(C) Yes, they merge into a single node

(D) Yes, same-level nodes are always directly linked

Correct Answer: B

Explanation: Level equality only reflects depth in the tree, not any direct relationship between the nodes.

Q75 | Application-based | Levels

In an organization chart with CEO -> Manager -> Employee, how many levels does this hierarchy have?

(A) Five levels

(B) Three levels

(C) One level

(D) Two levels

Correct Answer: B

Explanation: CEO, Manager, and Employee each occupy a distinct level, totaling three levels in this hierarchy.

3.1.10 Parent-Child Relationship

9 Questions

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Q76 | Definition-based | Parent-Child Relationship

The parent-child relationship in a Hierarchical Database refers to:

(A) A link only between two root nodes

(B) A link that changes direction randomly

(C) A link between two unrelated records at random levels

(D) The direct link between a higher-level record and its dependent lower-level record

Correct Answer: D

Explanation: This relationship directly connects a parent record to its subordinate child record.

Q77 | Feature-based | Parent-Child Relationship

In the parent-child relationship, which direction does the dependency generally flow?

A. From child to parent (parent depends on child)
B. No dependency exists
C. Both directions equally
D. From parent to child (child depends on parent)

Correct Answer: D

Explanation: Child records depend on their parent record for context and existence within the hierarchy.

Q78 | Statement-based | Parent-Child Relationship

Which of the following is TRUE about the parent-child relationship in this model?

A. A parent can have only one child, and a child can have many parents
B. Both parent and child can have unlimited connections both ways
C. Neither parent nor child can have more than one connection
D. A parent can have multiple children, but a child has only one parent

Correct Answer: D

Explanation: This asymmetric one-to-many rule is the defining feature of hierarchical parent-child links.

Q79 | Scenario-based | Parent-Child Relationship

A 'Department' record links to five 'Employee' records. What kind of relationship is this?

A. One-to-many parent-child relationship
B. One-to-one relationship
C. Many-to-many relationship
D. No relationship

Correct Answer: A

Explanation: One Department (parent) connecting to multiple Employees (children) is a classic one-to-many link.

Q80 | Concept-based | Parent-Child Relationship

Why is the parent-child relationship considered fundamental to the hierarchical model?

(A) It only applies to the root node

(B) It forms the basic building block that creates the entire tree structure

(C) It is used only for backup purposes

(D) It has no real impact on the database structure

Correct Answer: B

Explanation: Every branch of the hierarchical tree is built from repeated parent-child relationships.

Q81 | Assertion-Reason | Parent-Child Relationship

Assertion (A): A child record cannot exist independently of its parent record in a strict hierarchical database.
Reason (R): Access to child records is made through their parent record's path.
Choose the correct option:

(A) A is false, R is true

(B) Both A and R are true, and R correctly explains A

(C) Both A and R are false

(D) A is true, R is false

Correct Answer: B

Explanation: Since traversal paths run through the parent, a child's accessibility is tied to its parent record.

Q82 | Tricky | Parent-Child Relationship

If a child record needs to relate meaningfully to two different parent records, what problem arises in a Hierarchical Database?

(A) It cannot be represented directly, since each child allows only one parent

(B) The database automatically merges both parents

(C) It is represented easily with two direct links

(D) No issue arises at all

Correct Answer: A

Explanation: Representing two parents for one child is not possible under the strict single-parent hierarchical rule.

Q83 | Comparison-based | Parent-Child Relationship

How does the parent-child relationship in hierarchical databases compare to relationships in relational databases?

(A) Hierarchical relationships are physically embedded in the tree, while relational relationships are defined via keys

(B) Neither model defines any relationships

(C) Both use identical foreign key mechanisms

(D) Relational databases use tree links, not keys

Correct Answer: A

Explanation: Relational databases use foreign keys to link tables, whereas hierarchical databases embed the link structurally.

Q84 | Application-based | Parent-Child Relationship

In a library system, 'Author' is parent and 'Book' is child. If one author wrote many books, this reflects:

(A) A many-to-many relationship

(B) A relationship with no parent

(C) A valid one-to-many parent-child relationship

(D) An invalid relationship in this model

Correct Answer: C

Explanation: One Author linking to many Books directly matches the one-to-many parent-child pattern.

3.1.11 One-to-Many Relationship

8 Questions

OAVS TGT Computer Science — Hierarchical Database Syllabus

Q85 | Definition-based | One-to-Many Relationship

A one-to-many relationship in a Hierarchical Database means:

(A) No parent-child association exists

(B) Many parent records relate to one child record

(C) One parent record can be associated with several child records

(D) One parent record can relate to only one child record

Correct Answer: C

Explanation: This relationship type allows a single parent to be linked with multiple dependent children.

Q86 | Feature-based | One-to-Many Relationship

Which relationship type is exclusively supported between parent and child in the pure hierarchical model?

(A) Many-to-many only
(B) Zero-to-many only
(C) One-to-many only
(D) One-to-one only

Correct Answer: C

Explanation: The pure hierarchical model restricts parent-child links strictly to the one-to-many pattern.

Q87 | Statement-based | One-to-Many Relationship

Which of these examples correctly represents a one-to-many relationship?

(A) One Student linked to many Schools simultaneously
(B) One Class linked to many Students
(C) Many Teachers linked to many Subjects freely
(D) One Book linked to no author records

Correct Answer: B

Explanation: One Class connecting to many Students is a direct, valid one-to-many example.

Q88 | Scenario-based | One-to-Many Relationship

A 'Warehouse' record relates to hundreds of 'Item' records, but each Item belongs to only one Warehouse. This is:

(A) A many-to-many relationship
(B) An undefined relationship
(C) A one-to-one relationship
(D) A one-to-many relationship

Correct Answer: D

Explanation: Multiple Items tied to a single Warehouse, with each Item having only one Warehouse, is one-to-many.

Q89 | Concept-based | One-to-Many Relationship

Why can't a Hierarchical Database natively support many-to-many relationships?

A. Because it lacks any storage capacity
B. Because it only stores one record per table
C. Because each child is restricted to only one parent by design
D. Because it does not allow any relationships

Correct Answer: C

Explanation: Many-to-many needs multiple parents per child, which conflicts with the model's single-parent rule.

Q90 | Tricky | One-to-Many Relationship

To simulate a many-to-many relationship in a Hierarchical Database, what is typically required?

A. Converting all child nodes into root nodes
B. Duplicating child records under each relevant parent
C. Removing the parent-child link entirely
D. Nothing extra is required

Correct Answer: B

Explanation: Since true many-to-many isn't supported, data is often duplicated under multiple parents as a workaround.

Q91 | Application-based | One-to-Many Relationship

In a hospital database, one 'Doctor' record links to many 'Appointment' records. This models:

A. An unrelated pair of records
B. A one-to-many relationship between Doctor and Appointment
C. A one-to-one relationship
D. A many-to-one relationship from Appointment's side only, with no hierarchy

Correct Answer: B

Explanation: One Doctor associated with multiple Appointments is a standard one-to-many hierarchical example.

Q92 | Comparison-based | One-to-Many Relationship

How does a one-to-many relationship in Hierarchical Databases compare with relationships allowed in Network Databases?

A. Neither model supports one-to-many relationships

B. Hierarchical databases allow many-to-many but network databases do not

C. Both allow only one-to-one relationships

D. Network databases allow many-to-many, while hierarchical databases allow only one-to-many

Correct Answer: D

Explanation: Network databases relax the single-parent restriction, enabling many-to-many links unlike hierarchical databases.

3.1.12 Data Organization

9 Questions

OAVS TGT Computer Science — Hierarchical Database Syllabus

Q93 | Concept-based | Data Organization

In a Hierarchical Database, data organization follows which principle?

A. Alphabetical grouping only

B. Bottom-up grouping with no parent structure

C. Top-down grouping of related records into parent-child sets

D. Random distribution across unrelated files

Correct Answer: C

Explanation: Data is systematically grouped top-down, from general parent categories to specific child records.

Q94 | Feature-based | Data Organization

How is related data physically linked together in early hierarchical database implementations?

A. Using pointers connecting parent records to their child records

B. Using unrelated random file names

C. Using no linking mechanism at all

D. Using foreign keys between separate tables

Correct Answer: A

Explanation: Physical pointers were commonly used to connect parent records directly to child records.

Q95 | Statement-based | Data Organization

Which statement about the storage structure of Hierarchical Databases is TRUE?

(A) Child records are often stored physically close to or linked with their parent record

(B) All records are stored in a single unstructured blob

(C) Storage structure has no relation to the hierarchy

(D) Child records are always stored completely separately with no link

Correct Answer: A

Explanation: Physical storage in hierarchical databases commonly mirrors the logical parent-child structure.

Q96 | Basic | Data Organization

The overall storage approach in a Hierarchical Database is best described as:

(A) Based purely on random access memory addresses

(B) Structured and hierarchy-driven

(C) Completely unstructured

(D) Dependent only on cloud storage

Correct Answer: B

Explanation: Storage in hierarchical databases is organized to reflect the underlying tree hierarchy of data.

Q97 | Scenario-based | Data Organization

A company organizes its data as Company -> Branch -> Employee -> Salary Record. This demonstrates:

(A) Unrelated data scattered without structure

(B) A purely numeric storage scheme

(C) Flat, single-level data organization

(D) Hierarchical data organization with multiple nested levels

Correct Answer: D

Explanation: Multiple nested parent-child levels reflect the structured, hierarchical organization of data.

Q98 | Concept-based | Data Organization

What role does the 'segment' play in traditional hierarchical database storage structure?

A. It represents an index used only for backups
B. It has no defined role
C. It represents a network connection
D. It represents a record type or node at a given level of the hierarchy

Correct Answer: D

Explanation: A segment refers to a record type occupying a specific position within the hierarchical structure.

Q99 | Tricky | Data Organization

If child data needs to be retrieved without knowing its parent, what challenge exists in Hierarchical Databases?

A. The child data becomes the new root automatically
B. Retrieval is always instant regardless of parent
C. Direct retrieval is difficult since access typically requires traversing from the parent
D. No challenge exists at all

Correct Answer: C

Explanation: Hierarchical access paths generally require moving from root/parent downward, complicating independent child lookups.

Q100 | Application-based | Data Organization

An HR system stores Employee data nested under Department, nested under Company. This nested storage exemplifies:

A. Distributed ledger storage
B. Relational normalization
C. Object serialization only
D. Hierarchical data organization

Correct Answer: D

Explanation: Nested storage from Company down to Employee is a direct example of hierarchical organization.

Q101 | Comparison-based | Data Organization

How does hierarchical data organization differ from relational table-based organization?

(A) Neither model organizes data logically

(B) Both are identical in structure and storage

(C) Hierarchical organizes data as nested tree records; relational organizes data as independent linked tables

(D) Relational organizes data as a tree; hierarchical uses independent tables

Correct Answer: C

Explanation: Hierarchical nests records in a tree, whereas relational databases store data as separate linked tables.

3.1.13 Hierarchical Database Model

7 Questions

OAVS TGT Computer Science — Hierarchical Database Syllabus

Q102 | Concept-based | Hierarchical Database Model

The Hierarchical Database Model is best classified as a:

(A) Physical hardware model

(B) Logical data model based on tree structures

(C) User interface design model

(D) Networking protocol model

Correct Answer: B

Explanation: It is a logical model describing how data records relate, structured specifically as trees.

Q103 | Feature-based | Hierarchical Database Model

Which of the following models is considered a direct predecessor influencing later database models?

(A) Cloud Database Model

(B) Hierarchical Database Model

(C) In-memory Database Model

(D) Graph Database Model

Correct Answer: B

Explanation: The hierarchical model's concepts influenced the development of later models like the network and relational models.

Q104 | Statement-based | Hierarchical Database Model

Which statement about the Hierarchical Database Model is TRUE?

A. It has no defined relationship rules

B. It represents relationships using a strict tree hierarchy of records

C. It stores data without any parent-child linkage

D. It represents relationships using flexible many-to-many graphs

Correct Answer: B

Explanation: The model's core identity comes from its strict, tree-based representation of relationships.

Q105 | Comparison-based | Hierarchical Database Model

Which of these models is structurally most similar to the Hierarchical Database Model?

A. A fully connected mesh network diagram

B. A two-dimensional spreadsheet

C. A random unordered list

D. A single-rooted organizational tree diagram

Correct Answer: D

Explanation: The single-rooted tree diagram closely mirrors the hierarchical database's parent-child tree structure.

Q106 | Tricky | Hierarchical Database Model

Why is the Hierarchical Database Model sometimes described as 'less flexible' compared to later models?

A. Because it does not use records

B. Because relationships are restricted to a single parent per child

C. Because it supports unlimited relationship types

D. Because it has no structure at all

Correct Answer: B

Explanation: Flexibility is limited since only one-to-many, single-parent relationships can be modeled directly.

Q107 | Application-based | Hierarchical Database Model

A software team chooses a database model where each file directory contains sub-directories and files, but no sub-directory belongs to two parent directories. This matches the:

(A) Relational Database Model

(B) Network Database Model

(C) Hierarchical Database Model

(D) Object-Oriented Database Model

Correct Answer: C

Explanation: Restricting each sub-directory to a single parent directory reflects the hierarchical model's structure.

Q108 | Basic | Hierarchical Database Model

The Hierarchical Database Model primarily addresses which type of data relationship efficiently?

(A) Zero relationships

(B) One-to-many relationships

(C) Many-to-many relationships

(D) Circular relationships

Correct Answer: B

Explanation: This model is specifically well-suited to represent efficient one-to-many data relationships.

3.1.14 Navigation

7 Questions

OAVS TGT Computer Science — Hierarchical Database Syllabus

Q109 | Concept-based | Navigation

Navigation in a Hierarchical Database typically proceeds:

(A) Directly between unrelated nodes without paths

(B) From any random node in any direction

(C) From leaf nodes upward only, ignoring the root

(D) From the root node downward through parent-child links

Correct Answer: D

Explanation: Traversal conventionally starts at the root and follows defined parent-child paths downward.

Q110 | Feature-based | Navigation

Which term describes the specific route followed to access a child record from the root?

A. Foreign key path
B. Random walk path
C. Access path (or traversal path)
D. Index scan path

Correct Answer: C

Explanation: The access path defines the exact route through parent-child links used to reach a target record.

Q111 | Statement-based | Navigation

Which statement about navigation in a Hierarchical Database is TRUE?

A. Only leaf nodes can be navigated to directly
B. Any child record can be reached instantly without traversing its ancestors
C. To reach a deeply nested child, all its ancestor nodes must generally be traversed
D. Navigation is impossible in hierarchical databases

Correct Answer: C

Explanation: Because access follows the tree structure, reaching deep child nodes requires passing through ancestors.

Q112 | Scenario-based | Navigation

To find a specific 'Student' record under Department -> Class -> Student, the system must first access:

A. Department, then Class, then Student in sequence
B. Any record in random order
C. Class first, ignoring Department entirely
D. Only the Student record directly

Correct Answer: A

Explanation: Sequential traversal through Department and Class is required before reaching the Student record.

Q113 | Tricky | Navigation

What navigation challenge arises when trying to move from a child node directly to a 'sibling' child node?

(A) Siblings are always directly linked with no need for the parent

(B) There is often no direct sibling link; navigation must go back through the shared parent

(C) Sibling navigation is impossible under all circumstances

(D) Siblings automatically merge into one record

Correct Answer: B

Explanation: Since siblings share a parent but aren't directly linked, navigation typically routes back through that parent.

Q114 | Application-based | Navigation

In a hierarchical library database, to find all books by 'Author A', the system navigates:

(A) From an unrelated Publisher node with no connection

(B) Directly to a random Book node

(C) From the Book node upward to multiple unrelated authors

(D) From the Author node down to its linked Book child nodes

Correct Answer: D

Explanation: Finding an author's books means traversing from the Author (parent) down to its child Book records.

Q115 | Comparison-based | Navigation

How does navigation in a Hierarchical Database compare to navigation in a Relational Database?

(A) Relational databases only use tree paths, not queries

(B) Neither model supports structured navigation

(C) Both navigate using identical SQL joins

(D) Hierarchical navigation follows fixed tree paths, while relational navigation uses flexible queries via keys

Correct Answer: D

Explanation: Hierarchical databases rely on fixed paths, whereas relational databases use flexible key-based queries.

3.1.15 Advantages

9 Questions

Q116 | Feature-based | Advantages

Which of the following is a major advantage of the Hierarchical Database model?

(A) Ability to represent many-to-many relationships natively

(B) Complete independence from any structure

(C) Simplicity in representing one-to-many relationships

(D) Automatic support for unrelated data types

Correct Answer: C

Explanation: The model's simplicity in handling one-to-many links is one of its most cited advantages.

Q117 | Concept-based | Advantages

Why is data retrieval often fast in a well-designed Hierarchical Database?

(A) Because there is no need to define any paths

(B) Because predefined parent-child access paths reduce search time

(C) Because all data is stored without any structure

(D) Because retrieval always scans the entire database

Correct Answer: B

Explanation: Predefined paths let the system move directly along known links, speeding up retrieval.

Q118 | Statement-based | Advantages

Which of these is TRUE regarding the advantages of Hierarchical Databases?

(A) They perform poorly for any one-to-many data pattern

(B) They are only useful for many-to-many data

(C) They offer high performance for applications with clear one-to-many data patterns

(D) They offer no performance benefits over flat files

Correct Answer: C

Explanation: Hierarchical databases excel specifically where data naturally fits a one-to-many pattern.

Q119 | Comparison-based | Advantages

Compared to unstructured file storage, what advantage does a Hierarchical Database provide?

(A) Random data placement for security

(B) Slower access due to added complexity

(C) Complete removal of any organizational structure

(D) Clear, defined relationships that improve data integrity and organization

Correct Answer: D

Explanation: Defined parent-child relationships bring more organization and integrity than plain unstructured files.

Q120 | Scenario-based | Advantages

An organization with a strict single-manager reporting structure for employees would benefit from a Hierarchical Database because:

(A) It only works for flat organizations

(B) It cannot represent reporting structures at all

(C) The one-parent-per-child rule naturally matches the single-manager structure

(D) It requires multiple managers per employee

Correct Answer: C

Explanation: A single-manager hierarchy aligns perfectly with the model's one-parent-per-child restriction.

Q121 | Application-based | Advantages

Which application would benefit most from the Hierarchical Database's advantage of simple, fast one-to-many access?

(A) A recommendation engine linking many products to many users

(B) A file directory system with folders and files

(C) A collaborative document with multiple co-owners

(D) A social media app requiring many-to-many friend connections

Correct Answer: B

Explanation: Directory systems fit the one-to-many, single-parent pattern that hierarchical databases handle efficiently.

Q122 | Tricky | Advantages

Which advantage of the Hierarchical Database model becomes less useful as relationships grow more complex?

(A) Its simple one-to-many structure, since complex relationships need more flexibility

(B) Its lack of any structure

(C) Its support for unlimited parent connections

(D) Its ability to represent complex many-to-many links

Correct Answer: A

Explanation: The very simplicity that is an advantage for simple data becomes a limitation for complex relationships.

Q123 | Basic | Advantages

One advantage of the Hierarchical Database model related to data integrity is:

(A) It allows any data to be linked to any other data freely

(B) It enforces structural consistency through defined parent-child links

(C) It removes the need for any data validation

(D) It stores duplicate data with no restrictions

Correct Answer: B

Explanation: Enforced parent-child structure helps maintain consistent, predictable data relationships.

Q124 | Concept-based | Advantages

Why might a Hierarchical Database be considered efficient for read-heavy applications with stable structures?

(A) Because it constantly restructures data on every read

(B) Because it stores no actual data

(C) Because fixed access paths allow quick, predictable data retrieval

(D) Because it ignores all read requests

Correct Answer: C

Explanation: Stable, fixed paths mean read operations can follow known routes quickly without recalculating structure.

3.1.16 Limitations

9 Questions

OAVS TGT Computer Science — Hierarchical Database Syllabus

Q125 | Feature-based | Limitations

Which of the following is a significant limitation of the Hierarchical Database model?

A. Inability to store any data at all
B. Excessive flexibility in relationship types
C. Difficulty representing many-to-many relationships
D. Complete lack of parent-child structure

Correct Answer: C

Explanation: The strict one-parent rule makes representing many-to-many relationships difficult.

Q126 | Concept-based | Limitations

Why can modifying the structure of a Hierarchical Database be complex?

A. Because modifications are always instant with no impact
B. Because only leaf nodes exist
C. Because changing the hierarchy can affect all dependent child records
D. Because the database has no structure to modify

Correct Answer: C

Explanation: Since child records depend on their parent's position, structural changes can ripple through the hierarchy.

Q127 | Statement-based | Limitations

Which statement correctly identifies a limitation of the Hierarchical Database model?

A. It only works with unstructured data
B. It lacks native support for complex relationships beyond one-to-many
C. It supports every possible relationship type natively
D. It has no limitations compared to other models

Correct Answer: B

Explanation: A key documented limitation is its inability to natively model relationships beyond one-to-many.

Q128 | Comparison-based | Limitations

Compared to relational databases, a notable limitation of hierarchical databases is:

(A) Complete absence of any query capability

(B) Better support for many-to-many joins

(C) Reduced flexibility in querying data that doesn't follow the predefined hierarchy

(D) Superior flexibility for arbitrary ad-hoc queries

Correct Answer: C

Explanation: Relational databases allow flexible ad-hoc queries, while hierarchical databases are tied to fixed structures.

Q129 | Scenario-based | Limitations

A student needs to be linked to two different departments simultaneously in a Hierarchical Database. What limitation does this expose?

(A) The single-parent restriction makes this relationship difficult to represent directly

(B) This scenario is impossible to imagine

(C) This is easily represented with no issues

(D) The database automatically creates a new root for this case

Correct Answer: A

Explanation: Needing two parents for one child directly conflicts with the model's core single-parent limitation.

Q130 | Tricky | Limitations

Why is data redundancy sometimes unavoidable in Hierarchical Databases?

(A) Because redundancy is impossible in tree structures

(B) Because only one record can ever exist in the database

(C) Because the model forbids storing any duplicate data

(D) Because shared child data must be duplicated under each relevant parent

Correct Answer: D

Explanation: To associate the same data with multiple parents, duplication becomes necessary due to the single-parent rule.

Q131 | Application-based | Limitations

A hierarchical database models Employee -> Project, but an employee can work on multiple projects and vice versa. What limitation results?

(A) The many-to-many relationship cannot be modeled directly without redundancy or workarounds

(B) The relationship is modeled perfectly with no issues

(C) Employees are automatically restricted to one project only

(D) Projects disappear from the database

Correct Answer: A

Explanation: This many-to-many scenario clashes with the hierarchical model's native one-to-many limitation.

Q132 | Basic | Limitations

Which of these is generally considered a limitation regarding scalability in Hierarchical Databases?

(A) Scalability is never affected by hierarchy depth

(B) The model becomes simpler as data grows infinitely

(C) The model removes all levels automatically over time

(D) Complex hierarchies can become rigid and harder to maintain as data grows

Correct Answer: D

Explanation: As hierarchies deepen and grow, maintaining and modifying them can become increasingly complex.

Q133 | Assertion-Reason | Limitations

Assertion (A): Hierarchical Databases struggle with representing complex, interconnected data.
Reason (R): Each child record is restricted to a single parent record.
Choose the correct option:

(A) Both A and R are true, and R correctly explains A

(B) Both A and R are false

(C) A is false, R is true

(D) A is true, R is false

Correct Answer: A

Explanation: The single-parent restriction is exactly why complex, multi-linked data is hard to represent.

3.1.17 Applications

9 Questions

OAVS TGT Computer Science — Hierarchical Database Syllabus

Q134 | Application-based | Applications

Which of the following real-world systems is a classic example suited to the Hierarchical Database model?

(A) A ride-sharing matching system

(B) A social media friend-tagging system

(C) A peer-to-peer file sharing network

(D) A file system with folders and sub-folders

Correct Answer: D

Explanation: File systems naturally follow a single-parent folder structure, matching the hierarchical model well.

Q135 | Scenario-based | Applications

An organization chart showing CEO, Managers, and Employees with strict single-reporting lines is best modeled using:

(A) A Hierarchical Database

(B) A Network Database

(C) An unstructured flat file

(D) A Graph Database with cycles

Correct Answer: A

Explanation: Strict single-reporting lines map directly to the one-parent, one-to-many hierarchical structure.

Q136 | Basic | Applications

Which of these is historically an important application domain for hierarchical databases?

(A) Real-time multiplayer gaming graphics

(B) Banking and airline reservation record-keeping systems

(C) Social media image filters

(D) Web browser rendering engines

Correct Answer: B

Explanation: Early banking and airline systems used hierarchical databases like IBM IMS for structured record-keeping.

Q137 | Application-based | Applications

An XML document with nested tags (book > chapter > section) conceptually resembles which database model?

(A) Purely relational model

(B) Hierarchical Database model

(C) Purely network model

(D) Purely object-relational model

Correct Answer: B

Explanation: Nested XML tags reflect the same tree, single-parent structure found in hierarchical databases.

Q138 | Comparison-based | Applications

Why is a Hierarchical Database more suitable than a flat file for representing an organization's departmental structure?

(A) Neither model can represent departmental structures

(B) Flat files are always faster for nested data

(C) It naturally captures nested parent-child relationships between departments and sub-units

(D) Flat files naturally represent hierarchy better

Correct Answer: C

Explanation: Departmental nesting matches the hierarchical model's parent-child design far better than flat files.

Q139 | Tricky | Applications

Why might a Hierarchical Database be a poor choice for a modern social media friend-network application?

(A) Because social media has no data to store

(B) Because friend networks always fit a strict tree structure

(C) Because hierarchical databases cannot store any user data

(D) Because friend connections are many-to-many, which the model doesn't support natively

Correct Answer: D

Explanation: Friend networks require many-to-many relationships, which conflict with the hierarchical model's limitations.

Q140 | Statement-based | Applications

Which statement about applications of Hierarchical Databases is TRUE?

(A) They are exclusively used for image processing

(B) They are well suited to data with a clear, stable nested structure

(C) They are best suited for highly interconnected, non-hierarchical data

(D) They cannot be applied to any real-world system

Correct Answer: B

Explanation: Clear, stable nested data structures are the ideal use case for hierarchical databases.

Q141 | Application-based | Applications

A university's course catalog organized as Faculty -> Department -> Program -> Course is an example of applying the:

(A) Hierarchical Database model

(B) Purely distributed ledger model

(C) Purely in-memory caching model

(D) Purely event-driven model

Correct Answer: A

Explanation: This nested academic structure is a textbook application of the hierarchical database model.

Q142 | Scenario-based | Applications

A telecom company stores Region -> City -> Exchange -> Customer records. Which model fits this scenario best?

(A) A key-value cache only

(B) Flat file with no relationships

(C) A purely graph-based social model

(D) Hierarchical Database model

Correct Answer: D

Explanation: The nested Region-to-Customer structure is a natural fit for the hierarchical database model.

3.1.18 Comparison with File Organization

8 Questions

Q143 | Comparison-based | Comparison with File Organization

How does a Hierarchical Database improve upon traditional flat file organization?

(A) It introduces explicit parent-child relationships instead of isolated records

(B) It removes all relationships found in flat files

(C) It requires no records at all, unlike flat files

(D) It stores data with less structure than flat files

Correct Answer: A

Explanation: Flat files lack relationships between records, while hierarchical databases explicitly define them.

Q144 | Statement-based | Comparison with File Organization

Which of the following is TRUE when comparing Hierarchical Databases to traditional file systems?

(A) Hierarchical databases reduce data redundancy compared to fully duplicated flat files

(B) Hierarchical databases always require more physical storage regardless of design

(C) Flat files always have less redundancy than hierarchical databases

(D) Both systems are functionally identical

Correct Answer: A

Explanation: By linking related data through parent-child structures, hierarchical databases can reduce certain redundancies found in flat files.

Q145 | Concept-based | Comparison with File Organization

In traditional file organization, why is retrieving related data across files often harder than in a Hierarchical Database?

(A) Because hierarchical databases have no links at all

(B) Because file systems never store related data

(C) Because flat files typically lack built-in links between related records

(D) Because flat files always link every record automatically

Correct Answer: C

Explanation: Flat files generally require manual matching, whereas hierarchical databases provide direct parent-child links.

Q146 | Scenario-based | Comparison with File Organization

A company previously stored employee and department data in separate unrelated files, causing duplicate entries. Moving to a Hierarchical Database would primarily help by:

(A) Increasing the number of unrelated files further

(B) Establishing clear parent-child links between departments and employees

(C) Removing the need for any data storage

(D) Making department and employee data completely identical

Correct Answer: B

Explanation: Hierarchical structuring links departments (parent) and employees (child), reducing disconnected duplication.

Q147 | Tricky | Comparison with File Organization

Despite improvements over flat files, why do Hierarchical Databases still share one limitation with traditional file systems?

(A) Both eliminate the need for any records

(B) Both can struggle to efficiently represent complex many-to-many relationships

(C) Both automatically support unlimited many-to-many relationships

(D) Neither can store any structured data

Correct Answer: B

Explanation: Just like flat files, the hierarchical model isn't naturally suited for complex many-to-many data.

Q148 | Basic | Comparison with File Organization

Compared to traditional file organization, Hierarchical Databases provide better:

(A) Independence from any parent-child logic

(B) Structural organization through defined relationships

(C) Randomness in data placement

(D) Complete removal of structure

Correct Answer: B

Explanation: Defined parent-child relationships give hierarchical databases an organizational edge over flat files.

Q149 | Application-based | Comparison with File Organization

An old payroll system using separate flat files for 'Employee' and 'Salary' data is redesigned using a Hierarchical Database. What is the main introductory-level benefit?

(A) Salary records can now be directly linked as children of their respective Employee parent records

(B) Salary and Employee data become completely unrelated

(C) The system loses the ability to store salary data

(D) All employees are merged into a single record

Correct Answer: A

Explanation: Direct parent-child linkage between Employee and Salary improves organization over separate flat files.

Q150 | Feature-based | Comparison with File Organization

Which introductory-level feature makes Hierarchical Databases generally preferable to plain sequential flat files for nested data?

(A) Built-in support for representing nested one-to-many relationships

(B) Removal of the need to store any child data

(C) Complete lack of any file-like structure

(D) Requirement that all data be stored as a single string

Correct Answer: A

Explanation: Flat sequential files don't inherently support nested relationships the way hierarchical databases do.